The Neoclassical Approach to International Trade

1MT495 Current Issues in Applied International Economics

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Where We Are

Lectures 1–2: the accounting and the map

- The balance of payments records trade and its financing
- The *overall* trade balance is a macro outcome: $CA = S - I$
- Each item of the balance of payments has its own theory – today we take the goods and services lines

Today: the pattern of trade

- *Why* do countries export and import what they do?
- Who gains from trade – and who loses?

The models today assume **balanced trade** ($NX = 0$): they explain *what* a country trades, not its trade balance – that is saving and investment, from lectures 1–2.

Takeaway: Two questions, two tools: accounting tells us what must add up; trade theory tells us why the goods flow the way they do.

Outline

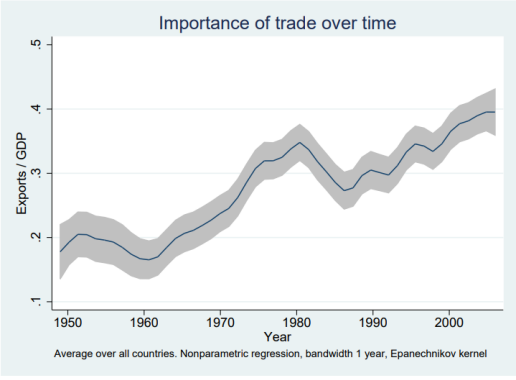
1. Introduction to Neoclassical Trade Theory
2. Empirical Evidence on Trade and Openness
3. The Ricardian Model
4. The Heckscher–Ohlin Model
5. Evidence and Distribution
6. Conclusions

Introduction to Neoclassical Trade Theory

- We start from the **foundations** of trade theory, building on the classical insights of Smith and Ricardo
- Neoclassical theory emphasises **efficiency**, **gains from trade** and the allocation of resources
- **Two key models:**
  - **Ricardian model:** trade is driven by differences in **relative productivity** (technology) – *comparative advantage*
  - **Heckscher–Ohlin (HO) model:** trade is driven by differences in **factor endowments** (capital, labour)
- **Standard assumptions:** perfect competition, no trade costs, balanced trade

Takeaway: Both models answer the same question – why trade? – with different sources of comparative advantage: technology versus endowments.

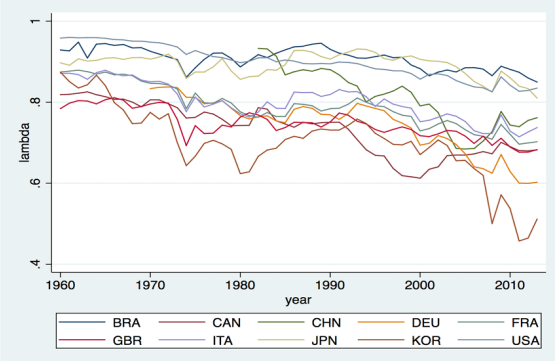
Trade Has Become More Important



Takeaway: The average country's exports rose from under 20% of GDP in the 1950s to about 40% by the late 2000s. For the Czech Republic the figure is 68.9% (2024).

Source: T. Allen, undergraduate international trade lecture notes; Czech figure: Eurostat (tet00003).

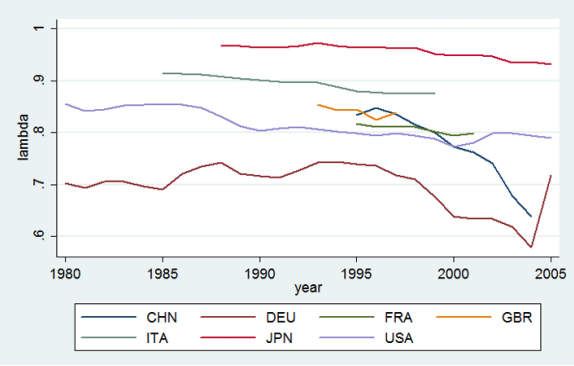
Openness to Trade: Share of Spending on Domestic Goods



Takeaway: λ is the share of spending on domestic goods ($1-$ the import share). It has fallen almost everywhere since 1960 – most in Korea (to about 0.5), least in the US and Brazil (still about 0.85–0.9).

Source: C. Arkolakis, "A Primer on Gains From Openness", lecture notes.

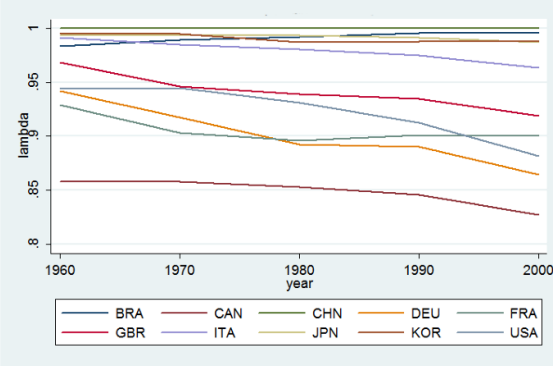
Openness to Multinationals: Domestic Share of Production



Takeaway: Here λ is the share of production by domestically owned firms. It falls as foreign multinationals produce locally – sharply in China in the early 2000s.

Source: C. Arkolakis, "A Primer on Gains From Openness", lecture notes.

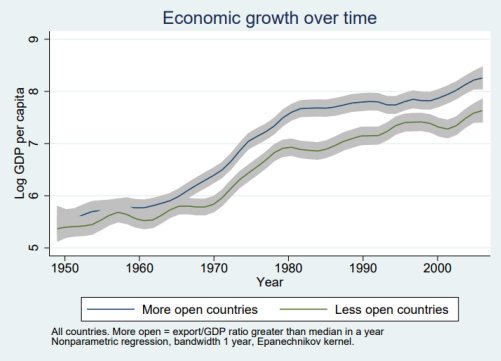
Openness to People: Native-Born Share of the Population



Takeaway: Here λ is the native-born share of the population. Migration has risen too, but much less than trade: in most countries λ is still above 0.9.

Source: C. Arkolakis, "A Primer on Gains From Openness", lecture notes.

Openness and Income



Takeaway: More open countries (exports/GDP above the median) are richer on average. This is a correlation: richer countries also trade more, so it does not by itself show that trade causes growth.

Source: T. Allen, undergraduate international trade lecture notes.

How Large Are the Gains From Trade?

A wide class of trade models – including Ricardian models with many goods (Eaton–Kortum) – delivers one formula (Arkolakis, Costinot and Rodríguez-Clare 2012):

$$\text{Loss from autarky (relative to trade)} = 1 - \lambda^{1/\varepsilon}$$

λ = share of spending on domestic goods; ε = trade elasticity (how strongly trade responds to trade costs), estimated at about 5–10.

United States, 2000

Import penetration 7% $\Rightarrow \lambda = 0.93$.
Autarky loss: **0.7% to 1.4%** of current real income.

A more open economy

$\lambda = 0.7$, $\varepsilon = 5$:
Autarky loss: $1 - 0.7^{0.2} \approx 6.9\%$ of current real income.

Takeaway: Two numbers – the domestic spending share and the trade elasticity – measure the welfare cost of closing to trade. The gain measured relative to autarky is $\lambda^{-1/\varepsilon} - 1$.

Source: Arkolakis, C., Costinot, A., & Rodríguez-Clare, A. (2012), American Economic Review, 102(1), 94–130.

The Ricardian Model: Main Ideas

- **Trade driver:** differences in **relative labour productivity** (technology) across countries
- **Core concept: comparative advantage**. A country exports the good it produces at a lower **opportunity cost** than other countries
- **Specialisation** follows comparative, not **absolute**, advantage: a country gains from trade even if it is less productive in every good
- **Gains from trade:** specialisation raises world output; trade lets each country consume beyond its own production possibilities

The Ricardian Model: Assumptions

- **Structure:** two countries, two goods, **labour (L) as the only factor of production**
- **Technology:** fixed **unit labour requirements** a_{Lj} – hours of labour per unit of good j, the inverse of productivity; constant returns to scale
- **Markets:** perfect competition
- **Mobility:** labour moves freely between sectors within a country but not across countries; all workers are identical

The Ricardian Model: Autarky and the PPF

- **Economic question:** what must the country give up to produce *one more unit of cloth*?
- **Scarce resource:** both sectors compete for the same fixed stock of labour:

$$a_{LW}Q_W + a_{LC}Q_C \leq L.$$

- **Opportunity cost:** one extra cloth uses a_{LC} hours. Those hours could have produced a_{LC}/a_{LW} units of wine.
- **Production possibility frontier:** with full employment,

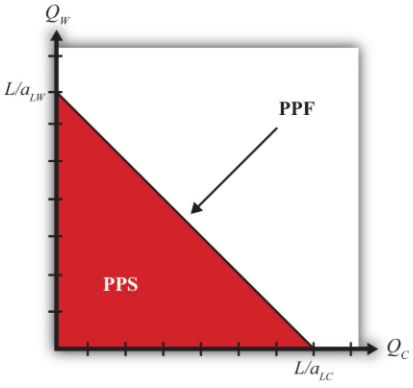
$$Q_W = \frac{L}{a_{LW}} - \underbrace{\frac{a_{LC}}{a_{LW}}}_{\text{wine sacrificed per cloth}} Q_C.$$

- **Autarky prices:** competition makes the relative price reflect this trade-off:

$$\boxed{\frac{p_C}{p_W} = \frac{a_{LC}}{a_{LW}}}.$$

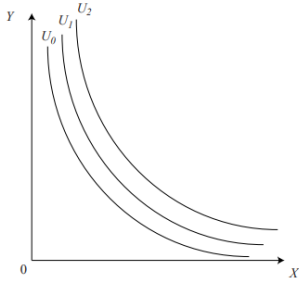
Takeaway: The PPF's slope is not just geometry: it is the real cost of moving a worker from wine to cloth.

The Ricardian Model: The PPF



- Intercepts: all labour in wine, L/a_{LW}; all labour in cloth, L/a_{LC}
- The shaded area is the production possibility set (PPS)
- The PPF is a **straight line** because there is one factor and constant returns: the opportunity cost does not change along the frontier

The Demand Side



On an indifference curve, $MRS = -\frac{dY}{dX} = \frac{\partial U/\partial X}{\partial U/\partial Y}$

At the consumer's optimum : $\frac{P_X}{P_Y} = MRS$

- Indifference curves $U_0 < U_1 < U_2$: combinations of the two goods with equal utility
- Along a curve: $MRS = -\dfrac{dY}{dX} = \dfrac{\partial U/\partial X}{\partial U/\partial Y}$
- At the consumer's optimum the price ratio equals the marginal rate of substitution: $\dfrac{p_X}{p_Y} = MRS$
- In autarky the country consumes where the highest attainable indifference curve touches its PPF

Ricardo's Example: England and Portugal

Unit labour requirements (hours per unit):

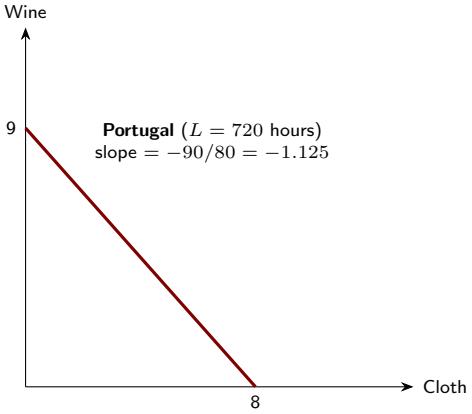
	Cloth	Wine
Portugal	90	80
England	100	120

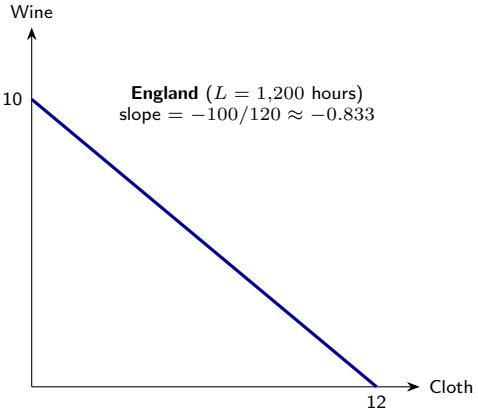
- Portugal has an **absolute advantage** in both goods
- Autarky price of cloth in terms of wine, $p_C/p_W = a_{LC}/a_{LW}$:
  - Portugal: $90/80 = 1.125$ – cloth is relatively expensive
  - England: $100/120 \approx 0.833$ – cloth is relatively cheap

Comparative advantage

- **Portugal in wine:** one unit of wine costs it $80/90 = 0.89$ units of cloth; in England, $120/100 = 1.2$
- **England in cloth:** one unit of cloth costs it 0.833 units of wine; in Portugal, 1.125

Ricardo's Example: The Two PPFs





The axes measure quantities, not labour hours. Each intercept divides available hours by hours per unit:

Portugal: $Q_C^{\max} = 720/90 = 8$, $Q_W^{\max} = 720/80 = 9$; England: $Q_C^{\max} = 1200/100 = 12$, $Q_W^{\max} = 1200/120 = 10$.

Takeaway: The *slope* identifies comparative advantage; the labour endowments also determine production capacities and world relative supply (Appendix A.2).

Trade Equilibrium and Specialisation

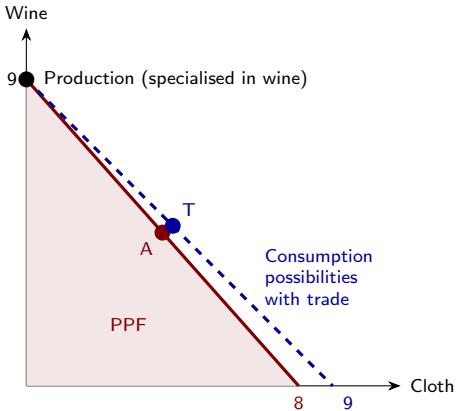
- **World price:** under complete specialisation, the relative cloth price lies **strictly between** the autarky prices

$$\underbrace{0.833}_{\text{England}} < \frac{p_C}{p_W} < \underbrace{1.125}_{\text{Portugal}}$$

- **Specialisation:** England makes cloth and Portugal wine. At either price bound, one country can instead produce both goods (Appendix A.3)
- **Ricardo's arithmetic** at a world price of 1 (one cloth for one wine):
  - Portugal gets a unit of cloth for 80 hours of work (making wine and trading it) instead of 90 – a saving of 11%
  - England gets a unit of wine for 100 hours instead of 120 – a saving of 17%

Takeaway: Both countries gain, even though Portugal is more productive in both goods: trade is an indirect, cheaper technology for producing the imported good.

Trade Separates Production From Consumption



Portugal at a world price $p_C/p_W = 1$

- **Production** moves to the corner of the PPF: Portugal makes only wine (9 units)
- **Consumption** lies on the line through that point with slope -1, the world price
- The line lies outside the PPF: Portugal can consume more of both goods (T instead of the autarky point A)

Takeaway: With trade, a country's consumption is no longer limited by its own PPF.

Wages and Prices

- **What determines a worker's wage?** One hour produces $1/a_{Lj}$ units of good j. Selling them earns p_j/a_{Lj} per hour.
- **Labour moves to the better-paying industry:**

$$w = \max\left\{\frac{p_C}{a_{LC}}, \frac{p_W}{a_{LW}}\right\}.$$

  The produced good satisfies $p_j = w a_{Lj}$; an inactive sector can have $p_j < w a_{Lj}$.
- **Why can Portugal pay higher wages?** At $p_C/p_W = 1$, Portugal makes wine and England makes cloth:

$$\frac{w_P}{w_E} = \frac{p_W/80}{p_C/100} = \frac{100}{80} = 1.25.$$

- **Range permitted by specialisation:** $100/90 \le w_P/w_E \le 120/80$, i.e. about 1.11 to 1.50.

Takeaway: Comparative advantage decides *which good* a country exports; absolute productivity helps explain *how much* its workers earn. Goods trade does not equalise wages.

The Ricardian Model: Implications and Extensions

- **Implications:**
  - Trade raises world output and both countries (weakly) gain
  - Trade is **not a zero-sum game**
  - With one factor and identical workers, trade produces no factor-based distributional conflict *within* a country
- **Criticisms:** too simple – one factor, complete specialisation, no income distribution, no trade costs, no trade between similar countries in similar goods
- **Modern extension – Eaton and Kortum (2002):**
  - A probabilistic Ricardian model with a continuum of goods and **geographic trade costs**
  - Countries buy each good from the cheapest supplier *after* trade costs – not necessarily the most productive one
  - It delivers a **gravity equation** and fits bilateral trade data well; it is the workhorse of quantitative trade policy analysis

The Heckscher–Ohlin Model: Context

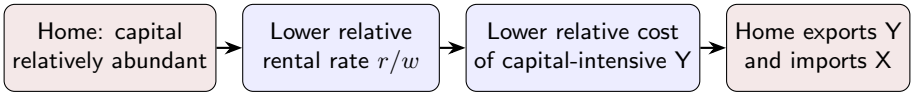
- **Historical motivation:** Heckscher (1919) and Ohlin (1933), formalised by Samuelson.
- **Ricardo asks:** why trade when countries have different technologies and only one factor of production?
- **Heckscher–Ohlin asks:** why trade even when countries have *the same technology*?
- **Answer:** a country with relatively abundant capital can produce capital-intensive goods at a relatively lower opportunity cost; a labour-abundant country has the opposite advantage.

Takeaway: The source of comparative advantage changes from productivity differences to differences in the relative scarcity of labour and capital.

Heckscher–Ohlin: The Economic Mechanism

Imagine two countries with identical technology

Home has relatively more **capital per worker**; Foreign has relatively more **workers per unit of capital**. Good Y uses capital intensively, good X uses labour intensively.



- **In autarky:** Y has a lower relative price at Home than abroad.
- **With trade:** world prices move between autarky prices, so Home expands Y and exports it.

Takeaway: Factor abundance matters because it changes *relative production costs* — not because a country must have more of every resource.

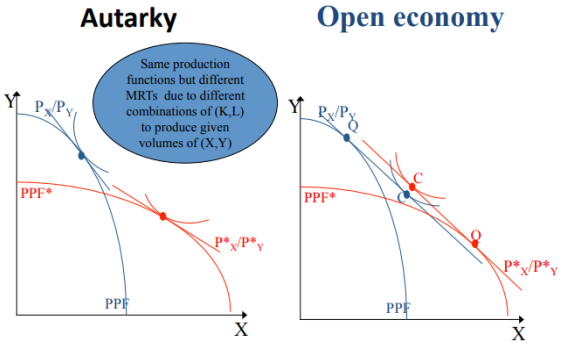
The Heckscher–Ohlin Model: Assumptions

- **Structure:** two countries (H, F), two goods (X, Y), two factors (L, K) – the $2 \times 2 \times 2$ model
- **Technology:** identical across countries, constant returns to scale
- **Factors:** mobile **within** countries, immobile **across** countries
- **Markets and preferences:** perfect competition; identical, homothetic preferences
- **No factor intensity reversals:** one good is labour-intensive relative to the other at every factor price

Model Setup

- Good X is **labour-intensive**: it uses more labour per unit of capital than Y at any factor prices
- Good Y is **capital-intensive**
- **Production functions:** $Q_j = Q_j(L_j, K_j)$ for $j = X, Y$, constant returns
- **Endowments:** $\bar{L}$ and $\bar{K}$ are fixed in each country
- **Full employment:** $\bar{L} = L_X + L_Y$ and $\bar{K} = K_X + K_Y$

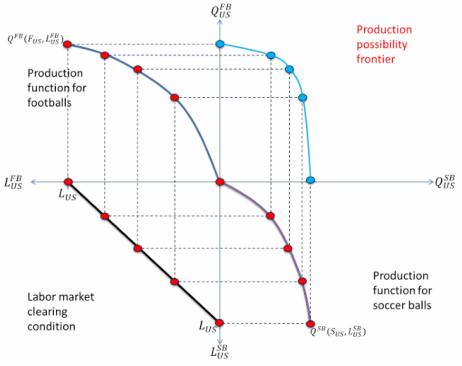
The Heckscher–Ohlin PPF



Unlike the Ricardian PPF, the HO PPF is **bowed out (concave)**. Because the two goods use the factors in *different proportions*, moving resources into one sector releases the wrong mix of factors and the opportunity cost rises along the frontier.

Takeaway: Opening to trade moves production along the PPF towards the good that uses the abundant factor intensively, and consumption to a higher indifference curve.

Deriving the PPF



Takeaway: The four-quadrant diagram maps the allocation of the factor across sectors (bottom left) through each sector's production function into the PPF (top right).

Source: T. Allen, undergraduate international trade lecture notes.

The Four Heckscher–Ohlin Theorems

- **Heckscher–Ohlin** (trade pattern): a country exports the good that uses its **relatively abundant factor** intensively
- **Factor price equalisation, FPE** (factor prices): free trade in goods can equalise factor prices (w, r) when both countries remain diversified
- **Stolper–Samuelson** (distribution): a rise in the relative price of a good raises the real return to the factor used intensively in it and lowers the real return to the other factor
- **Rybczynski** (production): at constant goods prices, an increase in one factor raises the output of the good that uses it intensively more than proportionally and *lowers* the output of the other good

Production and Unit Input Requirements

- **Economic question:** how many workers and machines does a firm need to make one unit — and how does this change when wages rise?
- **Technology and problem:** $Q_X = L_X^{\beta} K_X^{1-\beta}$; choose inputs to minimise $wL_X + rK_X$ subject to $Q_X = 1$, with $\beta = \theta_{XL} > \theta_{YL}$.
- **Firms substitute towards the cheaper factor:** the first-order conditions imply

$$\frac{K_X}{L_X} = \frac{1-\beta}{\beta}\frac{w}{r}.$$

  A higher w/r makes capital relatively attractive, raising K_X/L_X.
- **Convert the ratio into unit input requirements:** substitute $K_X = (K_X/L_X)L_X$ into the unit-output constraint (full algebra in A.4):

$$a_{LX} = \left(\frac{\beta}{1-\beta}\frac{r}{w}\right)^{1-\beta}, \qquad a_{KX} = \left(\frac{1-\beta}{\beta}\frac{w}{r}\right)^{\beta}.$$

Takeaway: The production technology is fixed, but the optimal inputs per unit *are not*: firms respond to relative wages and rental rates.

Equilibrium: The Matrix Structure

Two economic questions, answered by two linked systems:

1. What determines factor returns?

Firms cannot earn excess profits. Goods prices must cover the unit cost of labour and capital:

$$p_X = wa_{LX} + ra_{KX},$$
$$p_Y = wa_{LY} + ra_{KY}.$$

$$\boxed{\mathbf{p} = \mathbf{A}^\top \mathbf{w}}$$

2. What determines production?

The economy cannot use more factors than it has:

$$L = a_{LX}Q_X + a_{LY}Q_Y,$$
$$K = a_{KX}Q_X + a_{KY}Q_Y.$$

$$\boxed{\mathbf{V} = \mathbf{AQ}}$$

$$\mathbf{A} = \begin{pmatrix} a_{LX} & a_{LY} \\ a_{KX} & a_{KY} \end{pmatrix}, \quad \underbrace{\det \mathbf{A} = a_{LX}a_{KY} - a_{LY}a_{KX} > 0}_{\text{X is labour-intensive}}.$$

Takeaway: Changing *goods prices* moves factor returns (Stolper–Samuelson); changing *factor endowments* moves the output mix (Rybczynski).

Factor Price Equalisation

- **Intuition:** if the same two goods sell for the same prices in both countries, then firms with the same technologies must face the same unit-cost constraints.
- **Zero-profit equations:** for both goods to be produced,

$$p_X = c_X(w, r), \qquad p_Y = c_Y(w, r).$$

  The two equations jointly pin down the two unknown factor prices w and r (explicit Cobb–Douglas solution: A.5).
- **Required conditions:** identical technologies, common goods prices from free trade, no factor intensity reversals, and both countries remaining diversified (within their diversification cones).
- **Important qualification:** the input matrix $\mathbf{A}(w, r)$ changes with factor prices. We cannot treat it as constant when solving these price equations.

Takeaway: Trade can equalise wages and rental rates *without workers or capital crossing borders*: countries compete to produce goods at common world prices.

Stolper–Samuelson: Why Does the Wage Rise So Much?

Thought experiment: the price of labour-intensive X rises while the price of capital-intensive Y stays fixed.

1. **X becomes more profitable.** Firms in X want to expand, raising demand for workers and capital.
2. **Workers are used more intensively in X.** Wages are pushed up relative to capital's rental rate.
3. **But Y's price has not increased.** To keep producing Y at unchanged unit cost while wages rise, the rental rate on capital must fall.
4. **Magnification follows.** Wages must rise *more* than the price of X, while capital's rental rate falls: both sectors must still earn zero profit.

$$\boxed{\widehat{w} > \widehat{p}_X > \widehat{p}_Y = 0 > \widehat{r}}$$

Takeaway: Stolper–Samuelson is about the distribution of gains: the intensive factor gains purchasing power in terms of *both* goods, while the other factor loses.

Stolper–Samuelson: Cost Shares

- **What does a goods price contain?** Paying the workers and renting the machines used to make one unit.

$$p_j = w a_{Lj} + r a_{Kj}.$$

- **Cost shares** measure how much of each dollar of the price pays each factor:

$$\theta_{jL} = \frac{w a_{Lj}}{p_j}, \qquad \theta_{jK} = \frac{r a_{Kj}}{p_j}, \qquad \theta_{jL} + \theta_{jK} = 1.$$

- **Two sectors imply two different weights:** X is labour-intensive, so $\theta_{XL} > \theta_{YL}$. These weights determine how factor prices must adjust when goods prices change.

$$\boldsymbol{\Theta} = \begin{pmatrix} \theta_{XL} & \theta_{XK} \\ \theta_{YL} & \theta_{YK} \end{pmatrix}.$$

Takeaway: The matrix is economically meaningful: its first column shows how strongly each industry depends on wages; the second shows dependence on rental rates.

Stolper–Samuelson: The Derivation

- **Start from zero profit:** $p_X = w a_{LX} + r a_{KX}$. Differentiating includes both price and input adjustments:

$$dp_X = a_{LX}\,dw + a_{KX}\,dr + \underbrace{(w\,da_{LX} + r\,da_{KX})}_{=0 \text{ at the cost optimum}}.$$

- **Why does this last term disappear?** Firms already chose the least-cost input mix to produce one unit. A small reallocation along the unit isoquant has no first-order effect on total cost (envelope theorem).
- **Divide by price and apply the cost shares:**

$$\widehat{p}_X = \theta_{XL}\widehat{w} + \theta_{XK}\widehat{r}, \qquad \underbrace{\begin{pmatrix}\widehat{p}_X \\ \widehat{p}_Y\end{pmatrix}}_{\text{goods prices}} = \mathbf{\Theta}\ \underbrace{\begin{pmatrix}\widehat{w} \\ \widehat{r}\end{pmatrix}}_{\text{factor prices}}.$$

Takeaway: For small changes, each good's price change is a cost-share-weighted average of wage and rental-rate changes. The appendix solves this system step by step.

Stolper–Samuelson: The Magnification Effect

- **Invert:**

$$\begin{pmatrix}\hat{w}\\ \hat{r}\end{pmatrix} = \mathbf{\Theta}^{-1}\begin{pmatrix}\hat{p}_X\\ \hat{p}_Y\end{pmatrix}, \qquad \mathbf{\Theta}^{-1} = \frac{1}{\det\mathbf{\Theta}}\begin{pmatrix}\theta_{YK} & -\theta_{XK}\\ -\theta_{YL} & \theta_{XL}\end{pmatrix}, \qquad \det\mathbf{\Theta} = \theta_{XL}-\theta_{YL} > 0$$

- **Example:** p_X rises ($\hat{p}_X > 0$), p_Y constant, X labour-intensive:
  - $\hat{w} = \dfrac{\theta_{YK}}{\det\mathbf{\Theta}}\,\hat{p}_X > \hat{p}_X > 0$, because $\theta_{YK} = 1-\theta_{YL} > \theta_{XL}-\theta_{YL}$
  - $\hat{r} = -\dfrac{\theta_{YL}}{\det\mathbf{\Theta}}\,\hat{p}_X < 0$
- **First-order example:** $\theta_{XL} = 0.7$, $\theta_{YL} = 0.3$, $\hat{p}_X = 10\% \Rightarrow \hat{w} = 0.7\times 10/0.4 = 17.5\%$, $\hat{r} = -0.3\times 10/0.4 = -7.5\%$

Takeaway: Workers gain in terms of *both* goods ($\hat{w} > \hat{p}_X > \hat{p}_Y$); capital owners lose in terms of both. Trade creates winners and losers within a country.

Rybczynski Theorem: Derivation

- **Economic question:** suppose an economy receives more workers but no extra capital. Can both industries simply expand? **No**: each industry also needs capital.
- **Hold world goods prices fixed.** If both goods are produced, factor prices and optimal unit inputs $\mathbf{A}$ stay unchanged.
- **Apply full employment and invert:**

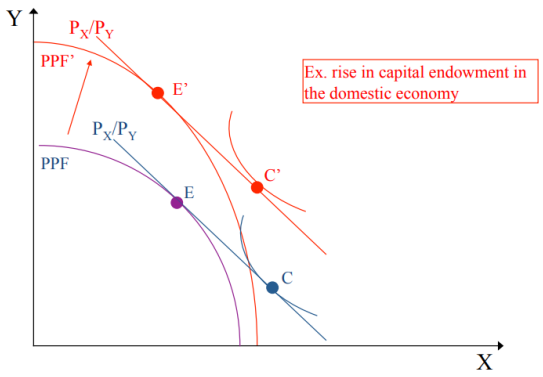
$$\underbrace{d\begin{pmatrix} Q_X \\ Q_Y \end{pmatrix}}_{\text{outputs}} = \mathbf{A}^{-1} d \underbrace{\begin{pmatrix} L \\ K \end{pmatrix}}_{\text{endowments}}, \quad \mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \begin{pmatrix} a_{KY} & -a_{LY} \\ -a_{KX} & a_{LX} \end{pmatrix}.$$

- **More labour, same capital** $(dL > 0,\ dK = 0)$:

$$\boxed{dQ_X = \frac{a_{KY}}{\det \mathbf{A}}\, dL > 0}, \quad \boxed{dQ_Y = -\frac{a_{KX}}{\det \mathbf{A}}\, dL < 0}.$$

Takeaway: The labour-intensive industry absorbs the extra workers, but to do so it must draw capital away from Y. This is why Y contracts *even though labour supply rises*.

Rybczynski Theorem: Graphically



- A rise in the capital endowment shifts the PPF outwards, **biased towards the capital-intensive good** (Y)
- At unchanged prices production moves from E to E′: more Y, less X

Summary of the Heckscher–Ohlin Theorems

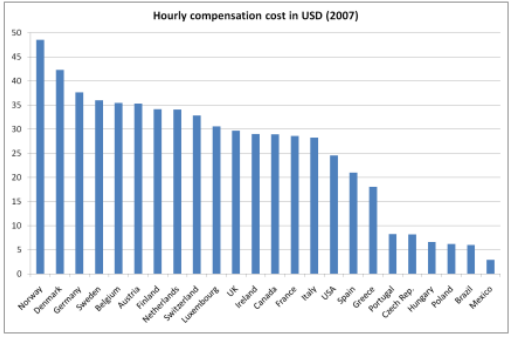
Theorem	**Driving force**	**Key result**
Heckscher–Ohlin	Endowment differences	Export the good intensive in the abundant factor
Factor price equalisation	Free trade in goods	w and r equalised across countries
Stolper–Samuelson	Change in goods prices	The intensive factor gains in real terms, the other loses
Rybczynski	Change in an endowment	The intensive good's output expands, the other contracts

The Heckscher–Ohlin Model Meets the Data

- **Successes:** predicts trade between countries with very different endowments (North–South trade) and the exports of resource-rich countries
- **Leontief paradox (1953):** US exports in 1947 were *less* capital-intensive than its imports, although the US was the most capital-abundant country
- **The "missing trade" (Trefler 1995):** the factor content of trade is far smaller than the model predicts
- **Repairs that work (Davis and Weinstein 2001):** allowing for technology differences, trade costs and non-traded goods brings predictions close to the data
- **Other limits:**
  - Identical technologies and FPE fail empirically
  - Much trade is **intra-industry** – cars for cars between similar countries
  - It works better with finer factors: skilled versus unskilled labour, land, resources

Takeaway: Endowments matter, but only together with technology differences and trade costs – which is why modern models combine Ricardo and HO.

No Wage Equalisation

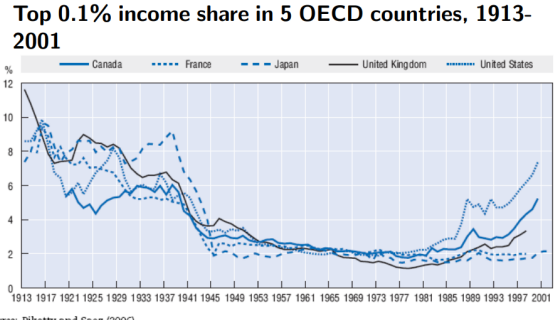


Source : US Bureau of Labor Statistics (August 2009)

Takeaway: Hourly compensation in manufacturing ranged from about $48 in Norway to about $3 in Mexico (2007). FPE fails: technologies differ, trade is costly and countries specialise.

Source: US Bureau of Labor Statistics (August 2009), via I. Méjean, lecture notes.

Trade and Inequality



Top 0.1% income share in five OECD countries, 1913–2001 (Piketty and Saez 2006)

- Top income shares fell until about 1980 and rose sharply in the US and the UK afterwards
- Trade is one candidate explanation – with technology, institutions and taxation
- **The China shock** (Autor, Dorn and Hanson 2013): Chinese import competition explains, conservatively, a quarter of the fall in US manufacturing employment in 1990–2007, with large, lasting local losses
- Workers do not move costlessly between sectors and regions, as the HO model assumes

Takeaway: Stolper–Samuelson predicts losers; the evidence shows they are concentrated in particular industries and places – the political economy of trade today.

Source: Piketty and Saez (2006), via I. Méjean, lecture notes; Autor, Dorn and Hanson (2013), American Economic Review, 103(6).

How the Neoclassical Models Relate

- **Ricardo:** trade driven by **technology differences**; one factor; typically complete specialisation
- **Heckscher–Ohlin:** trade driven by **endowment differences**; identical technologies; incomplete specialisation possible
- **Complements, not special cases:** HO with one factor would have no trade (technologies are identical); the **specific-factors model** bridges the two – factors stuck in their sector gain or lose with its price
- **Gains and distribution:** both models deliver **aggregate gains**
  - Ricardo: identical workers, so no losers within a country
  - HO: factor prices change, so there are **winners and losers** (Stolper–Samuelson)

Takeaway: Modern quantitative models (Eaton–Kortum with several factors) nest both sources of comparative advantage.

Conclusions

- **Neoclassical core:** trade yields aggregate gains through specialisation by comparative advantage – from technology (Ricardo) or endowments (HO)
- **Specialisation** lets countries consume beyond their own PPF; the gains can be measured with two numbers, λ and ε
- **Distribution:** trade changes factor prices, so some groups lose (Stolper–Samuelson); the China shock shows how concentrated the losses can be
- **Empirical limits:** identical technologies, FPE, balanced trade and perfect factor mobility fail in the data; intra-industry trade needs new theories (New Trade Theory, heterogeneous firms)
- **Modern relevance:** Ricardian models with trade costs (Eaton–Kortum) and extensions of HO remain the basis for explaining aggregate trade flows and distributional effects

Next lecture: new trade theory – why similar countries trade similar goods, and why only a few firms export.

Sources and Further Reading

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- Lecture notes: D. Donaldson (`https://dave-donaldson.com/teaching/`); T. Allen (`https://sites.google.com/site/treballen/undergraduate-trade/`); I. Méjean; G. Benigno (Background note on trade and trade imbalances); B. Shepherd (The gravity model of international trade); Saylor Academy, *International Trade: Theory and Policy*.

Appendix: Derivations and Worked Examples

What is the appendix for?

We first ask an **economic question**, then explain **which constraint matters**, show the **algebra**, and finally interpret the sign or magnitude.

- **A.1–A.3**: Ricardo — competitive prices, relative supply, world equilibrium and gains
- **A.4–A.7**: Heckscher–Ohlin — costs, factor-price effects and output effects

Takeaway: We reuse the main lecture's price, cost-share and factor-use matrices instead of introducing unrelated solution techniques.

Appendix Roadmap: What Changes, and What Stays Fixed?

Question	Shock	Held fixed	Response
Ricardo	Relative goods price	Technology, labour endowments	Specialisation and real wages
Stolper–Samuelson	Price of X rises	Technology, endowments	Wages up, capital returns down
Rybczynski	More labour available	Goods prices and unit inputs	More X, less Y

The organising principle

For Stolper–Samuelson, the *price system* transmits a shock into factor returns. For Rybczynski, the *full-employment system* transmits a shock into outputs. In both cases the proof is a 2×2 inversion.

A.1 Ricardo: Competitive Prices and Specialisation

Step 1 — unit cost. Producing good $j \in \{C, W\}$ uses a_{Lj} hours per unit at hourly wage w. Under perfect competition,

$$p_j \leq w a_{Lj}, \qquad p_j = w a_{Lj} \quad \text{if good } j \text{ is produced.}$$

Step 2 — mobile labour. A worker can earn p_j / a_{Lj} per hour in sector j. Therefore

$$w = \max\left\{ \frac{p_C}{a_{LC}}, \frac{p_W}{a_{LW}} \right\}.$$

Step 3 — autarky. If both goods are demanded, both must be produced, and the sectoral wage offers must coincide:

$$\frac{p_C}{p_W} = \frac{a_{LC}}{a_{LW}} \quad \text{(opportunity cost of cloth in wine units).}$$

Step 4 — trade. Write $p \equiv p_C / p_W$. Then

$$\begin{aligned} p > a_{LC}/a_{LW} \quad &\Rightarrow Q_C = L/a_{LC},\ Q_W = 0, \\ p < a_{LC}/a_{LW} \quad &\Rightarrow Q_C = 0,\ Q_W = L/a_{LW}. \end{aligned}$$

Takeaway: Price relative to domestic opportunity cost determines the export sector. At equality a country may diversify.

A.2 Ricardo: From Hours to Quantities (1/2)

Step 1 — distinguish inputs from output. In the main example, a_{Lj} is the *labour hours per unit* of good j, while L is a country's *available total hours*. At full specialisation, the PPF intercept is $Q_j^{\max} = L/a_{Lj}$.

	L (hours)	a_{LC}	a_{LW}	$Q_C^{\max} = L/a_{LC}$	$Q_W^{\max} = L/a_{LW}$
Portugal	720	90	80	$720/90 = \mathbf{8}$	$720/80 = \mathbf{9}$
England	1,200	100	120	$1200/100 = \mathbf{12}$	$1200/120 = \mathbf{10}$

Step 2 — turn the production decisions into world quantities. For a relative cloth price between $5/6$ and $9/8$, England makes 12 units of cloth and Portugal makes 9 units of wine:

$$RS \equiv \frac{Q_C^E + Q_C^P}{Q_W^E + Q_W^P} = \frac{12+0}{0+9} = \boxed{\frac{4}{3}}.$$

Takeaway: The numbers 8, 9, 10 and 12 are *output quantities*, obtained by dividing total labour hours by the corresponding hours-per-unit requirement.

A.2 Ricardo: World Relative Supply and Demand (2/2)

Using the capacities on A.2 (1/2), England can make 12 cloth and Portugal 9 wine. Their cloth opportunity costs are $p_E^A = 100/120 = 5/6$ and $p_P^A = 90/80 = 9/8$. Define $p = p_C/p_W$ and $RS = Q_C/Q_W$.

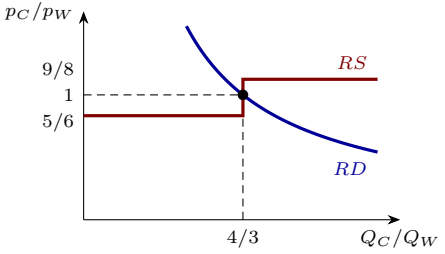
- $p < 5/6$: both countries make wine, so $RS = 0$.
- $5/6 < p < 9/8$: England makes cloth and Portugal wine:

$$RS = \frac{L^E/a_{LC}^E}{L^P/a_{LW}^P} = \frac{12}{9} = \frac{4}{3}.$$

- $p > 9/8$: both make cloth; $RS \to \infty$.
- At either threshold one country can produce both goods.

Demand. With Cobb–Douglas preferences and cloth expenditure share α, spending balance gives

$$RD(p) = \frac{C_C}{C_W} = \frac{\alpha}{1-\alpha}\frac{1}{p}.$$



At $\alpha = 4/7$ the intersection gives $p^W = 1$.

Takeaway: Complete specialisation fixes RS at $4/3$: in this diagram it is the *vertical* portion of relative supply.

A.3 Ricardo: Solving the England–Portugal Equilibrium

Production capacities. England: $L^E/a^E_{LC} = 1200/100 = 12$ cloth. Portugal: $L^P/a^P_{LW} = 720/80 = 9$ wine.

$$RD(p^W) = RS \quad \Rightarrow \quad \frac{\alpha}{1-\alpha}\frac{1}{p^W} = \frac{12}{9} \quad \Rightarrow \quad p^W = \frac{\alpha}{1-\alpha}\frac{9}{12}.$$

Case 1: $\alpha = 4/7$

$p^W = \frac{4}{3}\frac{9}{12} = 1$ lies strictly between $5/6$ and $9/8$. Thus England makes 12 cloth and Portugal 9 wine.

Cloth's share of world output value is $\frac{12}{12+9} = \frac{4}{7}$, as required.

Case 2: $\alpha = 1/2$

Interior formula gives $p^W = 3/4 < 5/6$ — inconsistent with England specialising in cloth.

Instead $p^W = 5/6$. World income in wine units is 19; cloth demand is $9.5/(5/6) = 11.4$. England produces 11.4 cloth and 0.5 wine.

Takeaway: A world price strictly between autarky prices implies complete specialisation; at an endpoint one country may diversify and receive no real-income gain.

A.4 Heckscher–Ohlin: Deriving Unit Inputs (1/2)

Intuition. Optimality fixes the *input ratio*; unit output fixes the *scale*.
Step 1 — cost minimisation. For one unit of good X, $L_X^{\beta} K_X^{1-\beta} = 1$ and $\beta = \theta_{XL} > \theta_{YL}$. Minimise $wL_X + rK_X$ subject to this constraint. **Step 2 — solve the first-order conditions.** The ratio of marginal products equals the ratio of factor prices:

$$\frac{w}{r} = \frac{\beta}{1-\beta}\frac{K_X}{L_X} \quad \Longrightarrow \quad \underbrace{\frac{K_X}{L_X}}_{A} = \frac{1-\beta}{\beta}\frac{w}{r}.$$

Step 3 — substitute back into the production constraint. Define $A \equiv \frac{1-\beta}{\beta}\frac{w}{r}$, so that $K_X = AL_X$. Then

$$\underbrace{L_X^{\beta}(AL_X)^{1-\beta}}_{\text{output}} = 1 \implies \underbrace{L_X^{\beta+1-\beta}}_{L_X} A^{1-\beta} = 1 \implies \boxed{L_X = A^{-(1-\beta)}}.$$

Step 4 — express labour per unit in factor prices.

$$\boxed{a_{LX} = \left(\frac{\beta}{1-\beta}\frac{r}{w}\right)^{1-\beta}}.$$

Takeaway: The key algebra is to substitute the optimal factor *ratio* into the unit-output constraint; only then do we obtain a unit input requirement.

A.4 Heckscher–Ohlin: Capital Inputs and Unit Cost (2/2)

Step 5 — recover the capital requirement. Since $K_X = AL_X$,

$$a_{KX} = A\,a_{LX} = A \cdot A^{-(1-\beta)} = A^{\beta} = \boxed{\left(\frac{1-\beta}{\beta}\frac{w}{r}\right)^{\beta}}.$$

Step 6 — find unit cost. At the optimal mix,

$$c_X(w,r) = w\,a_{LX} + r\,a_{KX} = \boxed{\frac{w^{\beta} r^{1-\beta}}{\beta^{\beta}(1-\beta)^{1-\beta}}}.$$

Step 7 — show why the exponents are cost shares. The FOCs imply $wL_X/(rK_X) = \beta/(1-\beta)$; therefore

$$\theta_{XL} = \frac{w\,a_{LX}}{c_X} = \beta, \qquad \theta_{XK} = \frac{r\,a_{KX}}{c_X} = 1-\beta.$$

The same calculation for Y replaces β by θ_{YL}.

Takeaway: The unit input coefficients $a_{ij}(w,r)$ are endogenous; for Cobb–Douglas, cost shares are fixed and equal the production exponents.

A.5 Heckscher–Ohlin: Factor Price Equalisation

Economic idea. At common world goods prices, two active industries cannot charge different unit costs across countries if they share the same technologies and an interior factor-price solution.

Step 1 — same price system as in the main lecture. Under constant returns and competition,

$$\mathbf{p} = \mathbf{A}(w, r)^{\top}\mathbf{w}, \qquad \mathbf{w} = \begin{pmatrix} w \\ r \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} a_{LX} & a_{LY} \\ a_{KX} & a_{KY} \end{pmatrix}.$$

$\mathbf{A}$ is *endogenous* because firms substitute factors; we must solve the unit-cost equations, not invert a fixed $\mathbf{A}$ while changing factor prices.

Step 2 — solve the Cobb–Douglas price system. For $j \in \{X, Y\}$, $p_j = \kappa_j w^{\theta_{jL}} r^{1-\theta_{jL}}$, $\kappa_j = [\theta_{jL}^{\theta_{jL}}(1-\theta_{jL})^{1-\theta_{jL}}]^{-1}$. Thus

$$\frac{p_X}{p_Y} = \frac{\kappa_X}{\kappa_Y}\left(\frac{w}{r}\right)^{\theta_{XL}-\theta_{YL}}, \quad \frac{w}{r} = \left(\frac{\kappa_Y p_X}{\kappa_X p_Y}\right)^{1/(\theta_{XL}-\theta_{YL})}.$$

Step 3 — recover levels. Let $q = w/r$; then $r = p_Y/(\kappa_Y q^{\theta_{YL}})$ and $w = qr$.

Takeaway: Same goods prices and technologies imply the same factor prices when both countries produce both goods (within the diversification cone).

A.6 Stolper–Samuelson: The Main Lecture's Matrix Method

Economic question. Why does a rise in the price of labour-intensive X raise wages *by more* than the product price itself? Because both industries must still break even.

Step 1 — envelope theorem. Write $p_j = c_j(w, r)$. Optimal factor demands obey

$$dp_j = a_{Lj}\, dw + a_{Kj}\, dr \quad \Rightarrow \quad \widehat{p}_j = \theta_{jL}\widehat{w} + \theta_{jK}\widehat{r}, \quad \theta_{jL} = \frac{w a_{Lj}}{p_j},\ \theta_{jK} = 1 - \theta_{jL}.$$

Step 2 — invert the same 2×2 system as in the main slides.

$$\begin{pmatrix} \widehat{p}_X \\ \widehat{p}_Y \end{pmatrix} = \underbrace{\begin{pmatrix} \theta_{XL} & \theta_{XK} \\ \theta_{YL} & \theta_{YK} \end{pmatrix}}_{\boldsymbol{\Theta}} \begin{pmatrix} \widehat{w} \\ \widehat{r} \end{pmatrix}, \quad \det \boldsymbol{\Theta} = \theta_{XL} - \theta_{YL} > 0.$$

$$\boxed{\widehat{w} = \frac{\theta_{YK}\widehat{p}_X - \theta_{XK}\widehat{p}_Y}{\theta_{XL} - \theta_{YL}}}, \quad \boxed{\widehat{r} = \frac{-\theta_{YL}\widehat{p}_X + \theta_{XL}\widehat{p}_Y}{\theta_{XL} - \theta_{YL}}}.$$

Step 3 — prove the real-return effects. For $\widehat{p}_X > \widehat{p}_Y$, since $\det \boldsymbol{\Theta} > 0$,

$$\widehat{w} - \widehat{p}_X = \frac{\theta_{XK}(\widehat{p}_X - \widehat{p}_Y)}{\det \boldsymbol{\Theta}} > 0, \quad \widehat{p}_Y - \widehat{r} = \frac{\theta_{YL}(\widehat{p}_X - \widehat{p}_Y)}{\det \boldsymbol{\Theta}} > 0.$$

Takeaway: If the relative price of labour-intensive X rises, $\widehat{w} > \widehat{p}_X > \widehat{p}_Y > \widehat{r}$: labour's real return rises and capital's falls.

A.7 Rybczynski: Invert the Employment System

Economic idea. Extra workers need capital to be productive. At unchanged factor prices, the economy reallocates capital from the capital-intensive industry to the labour-intensive one.

Step 1 — take the main lecture's factor-use system.

$$\mathbf{V} = \mathbf{A}\mathbf{Q}, \quad \begin{pmatrix} L \\ K \end{pmatrix} = \begin{pmatrix} a_{LX} & a_{LY} \\ a_{KX} & a_{KY} \end{pmatrix} \begin{pmatrix} Q_X \\ Q_Y \end{pmatrix}.$$

Step 2 — hold world goods prices fixed. FPE fixes w, r, hence firms' cost-minimising $\mathbf{A}$ stays unchanged, provided both sectors remain active.

$$d\mathbf{Q} = \mathbf{A}^{-1} d\mathbf{V}, \quad \mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \begin{pmatrix} a_{KY} & -a_{LY} \\ -a_{KX} & a_{LX} \end{pmatrix}, \quad \det \mathbf{A} > 0.$$

Step 3 — increase labour at fixed capital.

$$dK = 0,\ dL > 0 \quad \Rightarrow \quad \boxed{dQ_X = \frac{a_{KY}}{\det \mathbf{A}}\, dL > 0}, \quad \boxed{dQ_Y = -\frac{a_{KX}}{\det \mathbf{A}}\, dL < 0}.$$

Takeaway: The labour-intensive sector expands, the capital-intensive sector contracts: capital is reallocated to employ the extra workers.

A.7 (continued): Rybczynski Magnification in Percentages

Step 1 — convert units into employment shares. Define $\lambda_{ij} = a_{ij}Q_j/V_i$. Each factor's shares sum to one:

$$\begin{pmatrix} \widehat{L} \\ \widehat{K} \end{pmatrix} = \underbrace{\begin{pmatrix} \lambda_{LX} & \lambda_{LY} \\ \lambda_{KX} & \lambda_{KY} \end{pmatrix}}_{\mathbf{\Lambda}} \begin{pmatrix} \widehat{Q}_X \\ \widehat{Q}_Y \end{pmatrix}, \qquad \det \mathbf{\Lambda} = \lambda_{LX} - \lambda_{KX} > 0.$$

Step 2 — subtract, then solve.

$$\widehat{Q}_X - \widehat{Q}_Y = \frac{\widehat{L} - \widehat{K}}{\lambda_{LX} - \lambda_{KX}}, \quad \widehat{Q}_X = \widehat{L} + \lambda_{LY}(\widehat{Q}_X - \widehat{Q}_Y).$$

Step 3 — numerical check. If $\lambda_{LX} = 0.8$, $\lambda_{KX} = 0.4$, $\widehat{L} = 10\%$ and $\widehat{K} = 0$,

$$\widehat{Q}_X - \widehat{Q}_Y = 25\%, \quad \widehat{Q}_X = 15\%, \quad \widehat{Q}_Y = -10\%.$$

Check: both factor markets clear

$0.8(15) + 0.2(-10) = 10$ and $0.4(15) + 0.6(-10) = 0$.

Takeaway: The same matrix-inversion logic works for prices ($\mathbf{\Theta}^{-1}$) and for quantities ($\mathbf{\Lambda}^{-1}$): a larger factor endowment shifts output more than proportionately.